

Drilling Mechanics - APWD

Time Based

Real Time, Recorded Mode, Run 10

Schlumberger

Company: Equinor

Well: 31/5-7

Field: Eos

Rig Name: West Hercules

Country: Norway

Latitude: 60° 34' 35.14" N

UWID:

Longitude: 3° 26' 36.13" E

Rig Name:

West Hercules

Block: 31/5

Rig Type:

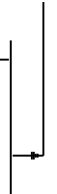
Semi-Submersible

FL: Norwegian North Sea

FL1: Section: 8 1/2 in.

FL2: Job no.: 19SCA0085

Log Measured From: - Drill Floor: 31.00 m
Permanent Datum: - Mean Sea Level



Ground Level: 307.00 m

Acquisition Dates: 31-Dec-2019 -- 01-Jan-2020

Other Services:

Log Interval: 12/31/2019 6:00:00 PM -- 1/1/2020 5:30:00 AM

Direction & Inclination (Surveys)

Index Types: Time

Directional Drilling

Index Scales: 5 cm / 3600 secs

geoVISION Bit Resistivity

Depth Source: Driller's Depth

VISION Resistivity

Depth Sensor: 3rd Party Depth

Print Type: Final

Spud Date: 02-Dec-2019



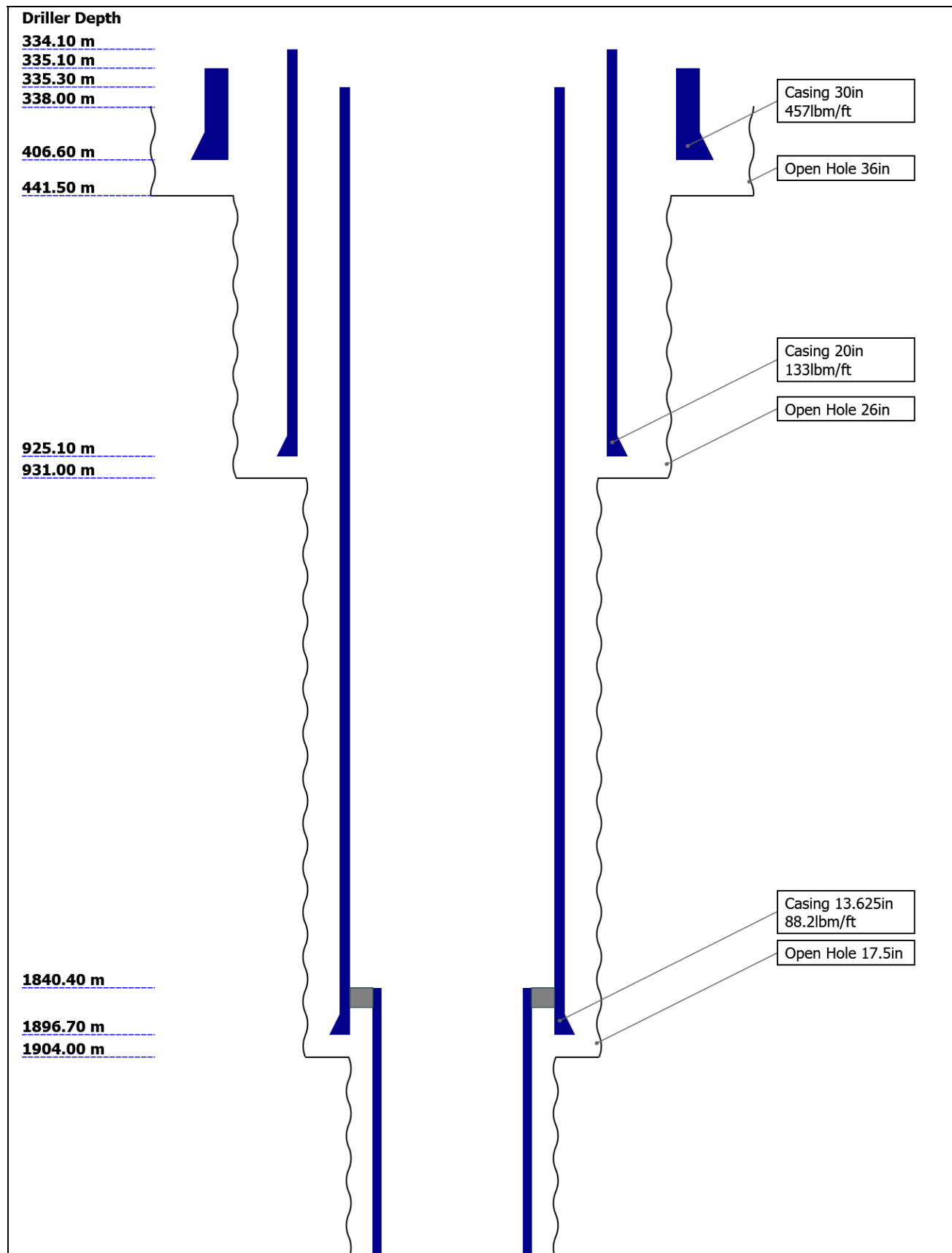
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Contents

- Header
- Disclaimer
- Contents
- Well Sketch
- Borehole Size/Casing/Tubing Record
- Operational Run Summary
- Borehole Fluids
- Remarks and Equipment Summary
- Run 10 Drilling Mechanics - APWD - Time Based
 - Integration Summary
 - Composite Summary
 - Log (Drilling Mechanics Light)
- Tail

Well Sketch



2583.00 m

2584.00 m

2643.00 m

Casing 9.625in

53.5lbm/ft

Open Hole 12.25in

Open Hole 8.5in

Borehole Size/Casing Record

Bit						
Bit Size (in)	36	26	17.5	12.25	8.5	
Top Driller (m)	338	441.5	931	1904	2584	
Bottom Driller (m)	441.5	931	1904	2584	2643	
Casing						
Size (in)	30	20	13.625	9.625		
Weight (lbm/ft)	457	133	88.2	53.5		
Inner Diameter (in)	27.067	18.73	12.375	8.535		
Grade	X56	N80	P110	P110		
Top Driller (m)	335.1	334.1	335.3	1840.4		
Bottom Driller (m)	406.6	925.1	1896.7	2583		

Operational Run Summary

Parameter (unit)	Run 10					
Date Log Started	31-Dec-2019					
Time Log Started	17:32:56					
Date Log Finished	01-Jan-2020					
Time Log Finished	07:38:34					
Bit Size (in)	8.500					
Bit Start Depth (m)	2596.00					
Bit Stop Depth (m)	2643.00					
Top Log Interval (m)	2585.90					
Bottom Log Interval (m)	2642.28					
Max Hole Deviation (deg)	1.12					
Azimuth of Max Deviation (deg)	120.28					
Logging Unit Number	A615					
Logging Unit Location	Service Office					
Recorded By	M.Martter					
Witnessed By	J.Eikenes					
Service Order Number	19SCA0085					

Borehole Fluids

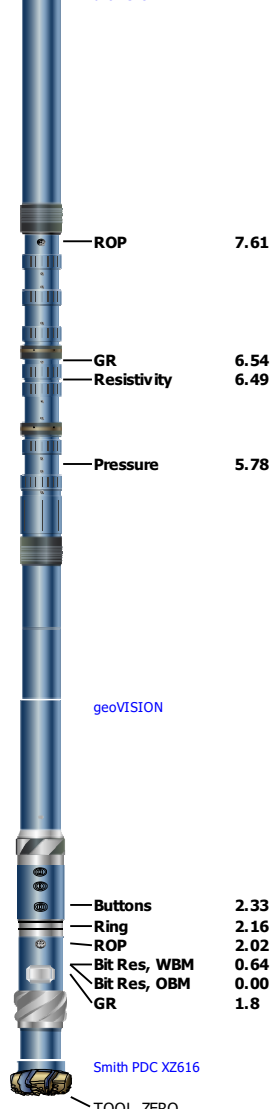
Parameter(unit)	Run 10					
Fluid Type	Water					
Fluid Name	Glydril					
Max Recorded Temperatures (degC)	43					
Source of Sample	Active Tank					
Salinity (ppm)	93984					
Density (g/cm3)	1.15					
Funnel Viscosity (s)						
Fluid Loss (cm3)						
PH						
Source RMF						
RMC	Pressed					
RM @ Meas Temp (ohm.m@degC)	0.07 @ 15					
RMF @ Meas Temp (ohm.m@degC)	0.06 @ 20					
RMC @ Meas Temp (ohm.m@degC)	0.09 @ 15					
RM @ BHT (ohm.m@degC)	0.04 @ 43					
RMF @ BHT (ohm.m@degC)	0.04 @ 43					
RMC @ BHT (ohm.m@degC)	0.05 @ 43					
Total Solid (%)						
High Gravity Solids (%)						

Remarks and Equipment Summary

Run 10: Toolstring				Run 10: Remarks	
Equip name TELE675:E4388 Length 18.37 MP name TeleScope Offset MSSU675:30848 Upper Extender MDC675:E4388 MMA:2507 MDI:3666 PMVC:236 PMEA:803 MTA:2162 MTK675:004 MSSD675:1005482-1 Lower Extender, Long 	All depths referenced to driller's depth. See EOWR for depth tracking details.				
	APWD data from tool memory. All other data is Real Time.				
	Gamma Ray measurement is environmentally corrected for mud weight, bit size, collar thickness and Potassium content.				
	Data acquired while reaming down over cored interval from 2592 to 2596 m bit depth.				
	Run 10 POOH at 2643 m to perform coring.				
	D&I 14.14				
	Vibration 13.14				
	ROP 11.79				

ARDC:H0143-1
Upper Extender:07/01
8
CDJA
AREA:618
APWD:283006
ARSS:915804-2
Lower Extender:14/04
5

RAB6:42913
RBDC:42913
Buttons:022
Stabilizer:81110D
Upper Extender:2853
RBBC
RBEC:216
Lower Extender



Lengths are in m
Maximum Outer Diameter = 8.500 in
Line: Sensor Location, Value: Gating Offset
All measurements are relative to TOOL_ZERO

Run 10

Drilling Mechanics - APWD - Time Based

Pass Summary

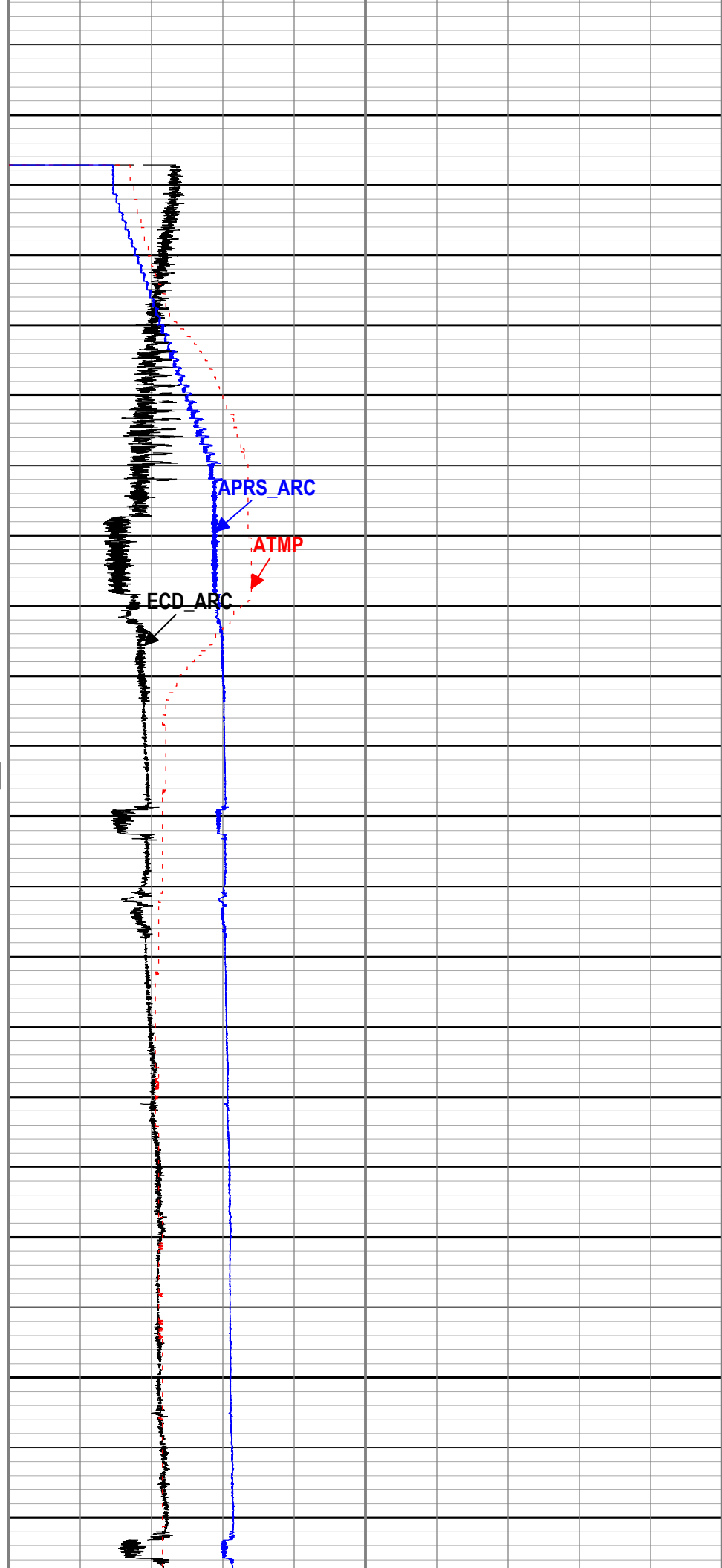
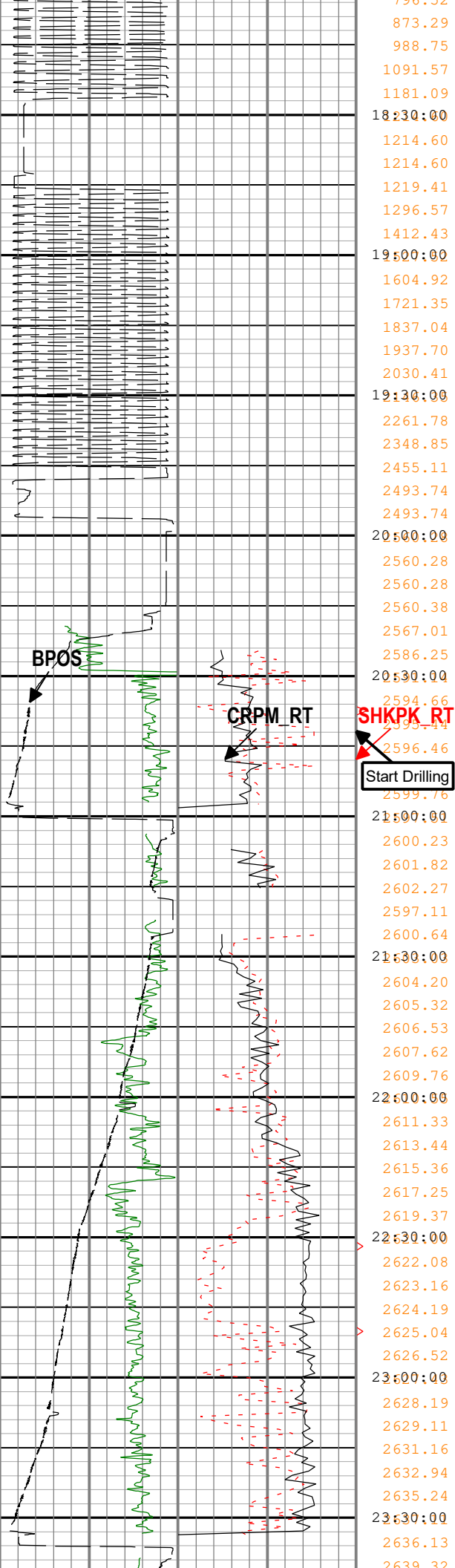
Run Name	Pass Objective	Direction	Start	Stop	Include Parallel Data
Run 10	TimeLogicalAcq	Down	31-Dec-2019 5:32:56 PM	01-Jan-2020 7:38:34 AM	Yes

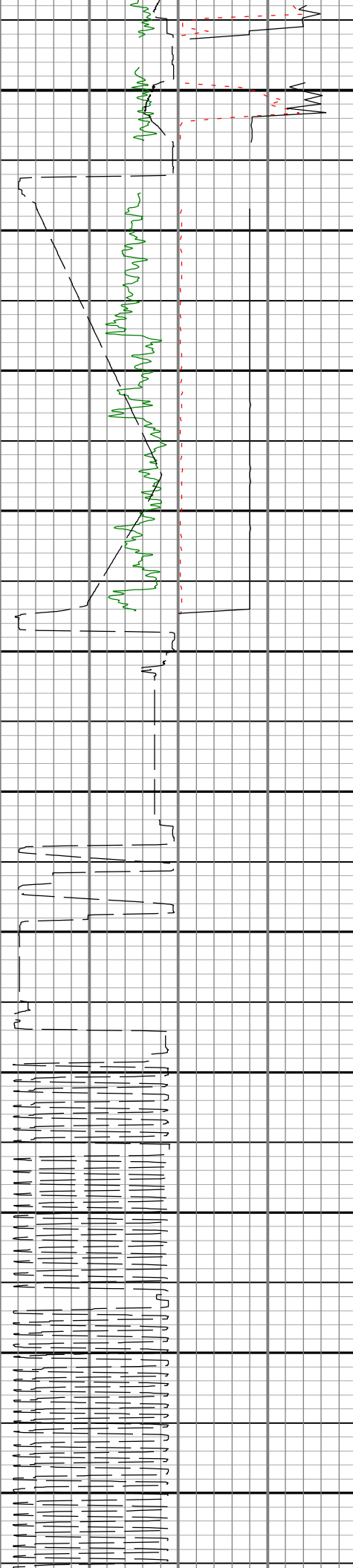
Log

Company:Equinor Well:31/5-7
Run 10: TimeLogicalAcq:S054

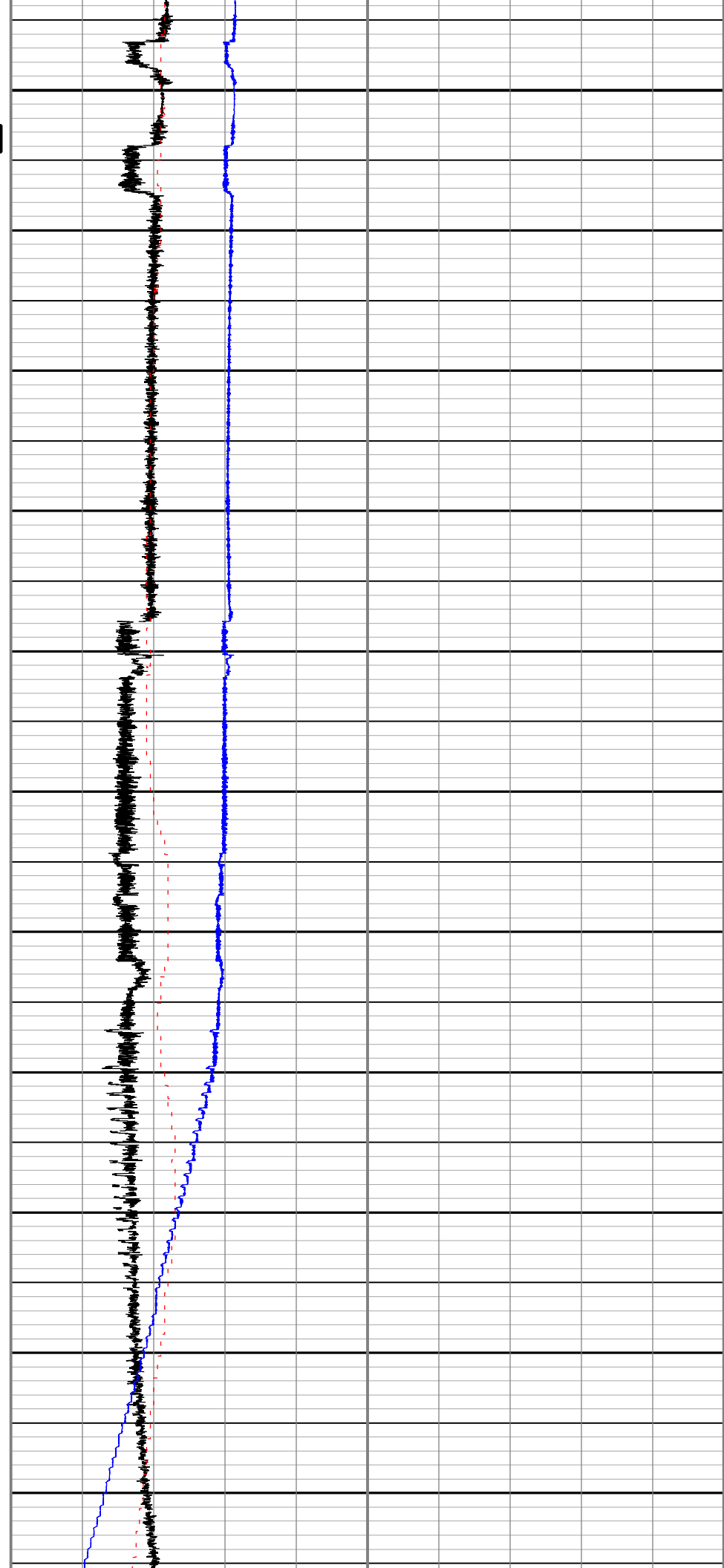
Description: Format: Log (Drilling Mechanics Light) Index Scale: 5 cm per 3600 s Index Type: Time Creation Date: 21-Jan-2020 10:33:22

		Depth Index (DEPTH) RT m											
Gamma Ray (GR_ARC_RT) ARC6 RT	Stick Slip Indicator (STICK_RT) TELE675 RT		Shock Risk (SHKRSK_R T) TELE675 RT	Equivalent Circulating Density (ECD_ARC) ARC6 RM									
	0 c/min 500		5 0	1 g/cm3 2									
Height of block above rig floor (BPOS) RT	Collar Rotational Speed (CRPM_RT) TELE675 RT		Shock Peak (SHKPK_RT) TELE675 RT	Downhole Annulus Temperature (ATMP) ARC6 RM									
	0 m 45		0 gn 250	0 degC 200									
				Downhole Annulus Pressure (APRS_ARC) ARC6 RM									
				0 bar 1000									
			18:00										





2639.92
2637.74
2636.35
2636.11
Jan, 01 00:00
2641.38
2641.74
Stop Drilling
2635.84
2635.84
2630.94
00:30:00
2625.38
2622.61
2619.82
2617.05
2614.28
01:00:00
2608.73
2605.96
2603.19
2600.43
2601.28
01:30:00
2608.02
2611.60
2615.17
2619.61
2635.53
02:00:00
2639.85
2639.74
2639.74
2639.74
2639.74
02:30:00
2639.74
2635.10
2595.87
2595.96
2556.69
03:00:00
2556.24
2556.27
2556.33
2556.33
2517.75
03:30:00
2401.04
2323.50
2245.23
2207.63
2129.68
04:00:00
1973.66
1895.47
1817.69
1778.67
1700.70
04:30:00
1544.82
1465.35
1349.51
1272.62
1194.38
05:00:00
1038.77
961.43
884.27



Gamma Ray (GR_ARC_RT) ARC6 RT 0gAPI150 Height of block above rig floor (BPOS) RT 0m45			Stick Slip Indicator (STICK_RT) TELE675 RT 0c/min500 Collar Rotational Speed (CRPM_RT) TELE675 RT 0c/min250			Depth Index (DEPTH) RT m Shock Risk (SHKRSK_RT) TELE675 RT 50 Shock Peak (SHKPK_RT) TELE675 RT 0gn250			Equivalent Circulating Density (ECD_ARC) ARC6 RM 1g/cm32 Downhole Annulus Temperature (ATMP) ARC6 RM 0degC200 Downhole Annulus Pressure (APRS_ARC) ARC6 RM 0bar1000		
Description: Format: Log (Drilling Mechanics Light) Index Scale: 5 cm per 3600 s Index Type: Time Creation Date: 21-Jan-2020 10:33:22											

Company:

Equinor

Well:

31/5-7

Field:

Eos

Rig Name:

West Hercules

Country:

Norway



Schlumberger

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Time Based

Real Time, Recorded Mode, Run 10